MICE Structure Map for the Introduction

From the current situation to the thesis surrogate workflow

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MICE Model: Four Moves in an Engineering Thesis Introduction

Move	Function	Question answered in this thesis
Move 1	Current situation	Why is pilot spray in multi-fuel marine engines worth studying?
Move 2	Problems	What is missing from experiments, CFD, empirical models, and image measurements?
Move 3	New approach	Why is learning a surrogate from Mie videos a reasonable route?
Move 4	Your solution	What does this thesis do, how does it do it, and what is its scope?

The MICE model emphasizes a general-to-specific and problem-solution order. This deck uses it to audit the argument chain in the introduction.

Overall Logic of the Introduction

1. Industry background: shipping decarbonization and alternative fuels drive multi-fuel marine-engine development.
2. Technical background: many multi-fuel engines still need pilot fuel for reliable ignition.
3. Engineering risk: pilot diesel spray development may lead to impingement, lubricant dilution, and deteriorated combustion.
4. Method gap: engine tests and CFD are costly, while image measurements alone do not form a fast predictive screening model.
5. New route: build reduced-order penetration targets from high-speed Mie imaging and train an uncertainty-aware surrogate.

Move 1-1: Making a Centrality Claim

Move 1: Current situation

Step 1: Making a centrality claim

Thesis argument

International shipping faces simultaneous pressure to reduce greenhouse-gas emissions, diversify fuels, and maintain operational reliability. The IMO 2023 strategy and industry orderbook trends make alternative-fuel propulsion a practical engineering issue [imo2023ghgstrategy; classnk2025alternativefuels].

- Function: places the work in the wider context of marine energy transition.
- Keywords: importance, relevance, application.
- Introduction location: Research relevance, paragraph 1.

Move 1-2: Making Topic Generalizations

Move 1: Current situation

Step 2: Making topic generalizations

Thesis argument

Dual-fuel and multi-fuel marine engines require not only alternative-fuel supply but also reliable ignition. Low-pressure gas, methanol, and ammonia concepts commonly still require a small quantity of pilot fuel [wingd_xdf_manual_2021; wingd_lng_faq_2025; classnk_methanol_2024; wingd_ammonia_faq_2025].

- Function: explains the general role of pilot fuel in the engine system.
- Narrowing step: pilot diesel spray is not only an ignition aid; its spatial development also creates engineering risk.
- Introduction location: Research relevance, paragraphs 2–3.

Move 1-3: Reviewing Previous Research and Solutions

Move 1: Current situation

Step 3: Reviewing previous research and solutions

Thesis argument

Spray behavior and impingement risk are commonly assessed through engine tests, CFD and spray-combustion simulations, and optical spray experiments. Diesel-spray research already has a foundation in high-speed imaging, empirical correlations, phenomenological models, and CFD [Linne2013SprayImaging; HiroyasuArai1990; NaberSiebers1996; liu2020sprayreview].

- Function: acknowledges existing work instead of claiming that diesel spray is unstudied.
- Introduction location: Current engineering practice and Existing solutions.

Move 2-1: Indicating Weaknesses

Move 2: Problems

Step 1: Indicating weaknesses

- Engine tests are closest to real operation, but they are expensive, slow, and hard to use for isolating one design factor.
- High-fidelity CFD is information-rich, but it requires complex models, many assumptions, and substantial computational resources.
- One reacting diesel-spray LES example required about 48,000 CPU core-hours and up to 20 days of wall-clock time for only 2 ms after SOI [CurtisNiemeyerSung2017].
- Empirical and phenomenological models are compact and interpretable, but less robust for short-pulse, strongly transient, and multiple-injection cases.
- Optical boundaries are method-dependent experimental definitions, not hidden physical truth fields [ecn_liquid_penetration; ecn_liquid_penetration_accessory].

Move 2-2: Identifying a Gap

Move 2: Problems

Step 2: Identifying a gap

Thesis gap statement

What is missing is a fast and repeatable method for predicting the time-evolving non-reacting liquid envelope of pilot diesel spray, so that later impingement-risk analysis has a defensible geometric input.

- The gap is not that diesel spray has never been studied.
- The missing element is an intermediate-layer predictor between optical experiments and high-cost validation.
- The aim is not to replace CFD or engine tests, but to narrow the candidate space before using them.

Move 3-1: Introducing a Potential Solution

Move 3: Proposing a new approach

Step 1: Introducing a potential solution

Thesis route

Learn a condition-parametric surrogate directly from large-scale Mie-scattering optical spray data, turning measured penetration trajectories into a fast predictive model.

- Mie videos already capture liquid-envelope behavior.
- Reduced-order fitting converts raw observations into stable learning targets.
- The surrogate sits between raw measurement and physics-based simulation.

Move 3-2: Justifying the Approach

Move 3: Proposing a new approach

Step 2: Justifying the approach

- Experimental fidelity: learning targets come from actual Mie images and penetration trajectories.
- Screening speed: avoids sending every candidate condition directly to CFD or engine testing.
- Data extensibility: the condition-to-trajectory-parameter mapping can be updated as the dataset grows.
- Interpretability and auditability: calibration, boundary definition, trajectory fitting, and uncertainty regression all leave intermediate artifacts.
- Consistency with scientific imaging practice: Fiji/ImageJ, scikit-image, OpenPIV, and PIVlab emphasize scriptable, inspectable, reusable pipelines [Schindelin2012Fiji; VanDerWalt2014ScikitImage; OpenPIV; ThielickeSonntag2021PIVlab].

Move 4-1: Research Aims

Move 4: Your solution

Step 1: Research aims

Thesis aim

Develop a data-driven non-reacting liquid-envelope prediction model for pilot diesel spray. The model predicts spray-envelope and penetration trends from injection-condition and time information, together with uncertainty estimates suitable for engineering interpretation.

- Output: mean penetration trend plus uncertainty.
- Use: early-stage impingement-risk screening in marine dual-fuel engine development.

Move 4-2: Methods

Move 4: Your solution

Step 2: Methods

1. High-speed Mie imaging of diesel spray in a room-temperature pressurized-air chamber.
2. Manual calibration of nozzle center and pixel scale.
3. CUDA-accelerated batch processing to extract plume-resolved temporal observables and boundary geometry.
4. Reduced-order fitting to construct penetration targets and handle right-censoring and temporal alignment.
5. Multi-stage learning: Stage-1 MSE, Stage-2 heteroscedastic NLL, and teacher-guided refinement.
6. Coupling predicted mean and uncertainty to simplified geometric risk screening.

Move 4-3: Thesis Scope

Move 4: Your solution

Step 3: Thesis scope

- In scope: controlled pressurized-chamber, room-temperature, non-reacting, non-evaporating pilot diesel spray.
- In scope: macroscopic liquid-envelope geometry and time evolution as observed in the optical setup.
- Out of scope: combustion chemistry, evaporation, dual-fuel ignition coupling, real hot-wall interaction, and real-engine wall-film truth.
- Model boundary: mainly interpolation within the covered data domain; clear OOD extrapolation is not guaranteed at the same level.

Move 4-4: Main Outcome

Move 4: Your solution

Step 4: Describing the main outcome

1. A CUDA-accelerated workflow from terabyte-scale Mie videos to plume-resolved observables.
2. Reduced-order fitting for trajectory alignment, robust extrapolation, and mitigation of right-censoring.
3. A multi-stage surrogate combining MSE, heteroscedastic NLL, and teacher-guided refinement.
4. A probabilistic screening prototype that decouples penetration prediction from downstream geometric collision evaluation.

Move 4-5: Thesis Structure

Move 4: Your solution

Step 5: Overview of thesis structure

Part	Function
Literature review and technical rationale	Situates the work in diesel spray imaging, optical diagnostics, and surrogate modeling.
Materials and methods	Describes the experimental data, image processing, feature extraction, trajectory fitting, and model training.
Results	Reports archived processing assets and model evidence.
Conclusions	Summarizes the outcome, limitations, and remaining development needs.

Move/Step Checklist

MICE position	Introduction subsection	Core argument
Move 1-1	Research relevance	Shipping decarbonization and alternative fuels make the topic important.
Move 1-2	Research relevance	Multi-fuel engines rely on pilot fuel, and spray spatial development creates risk.
Move 1-3	Current practice / Existing solutions	Experiments, CFD, optical diagnostics, and empirical models already exist.
Move 2-1	Existing solutions	Existing methods have weaknesses in cost, transience, interpretability, or truth definition.
Move 2-2	Problem statement	A fast intermediate-layer predictor is missing between optical experiments and high-cost validation.
Move 3-1	End of problem statement	Learn a surrogate from Mie data.
Move 3-2	Why this workflow is necessary	Reduced targets, auditability, uncertainty, and high-throughput processing justify the route.
Move 4-1–5	Aim / Methods / Scope / Contributions / Structure	The thesis solution, boundary, outcomes, and organization.

Source Keys

This deck uses citation keys from the thesis bibliography as lightweight source markers. The formal references remain in the thesis PDF bibliography.

Topic	Source keys
Shipping and alternative fuels	`imo2023ghgstrategy`; `classnk2025alternativefuels`
Pilot-fuel background	`wingd_xdf_manual_2021`; `wingd_lng_faq_2025`; `classnk_methanol_2024`; `wingd_ammonia_faq_2025`
Spray experiments and models	`Linne2013SprayImaging`; `HiroyasuArai1990`; `NaberSiebers1996`; `liu2020sprayreview`
CFD cost	`CurtisNiemeyerSung2017`
Optical boundary definition	`ecn_liquid_penetration`; `ecn_liquid_penetration_accessory`
Image-processing ecosystems	`Schindelin2012Fiji`; `VanDerWalt2014ScikitImage`; `OpenPIV`; `ThielickeSonntag2021PIVlab`